

Automobile Chassis

Complete student lesson for course code **4411**.

Revision 2026 Semester 4 Program Core 4 modules

Course snapshot

Scheme: 4 (L:3,T:1,P:0); 60 periods

Credits: 4

Contact: 60 periods per semester

Module hours listed: 58

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

4411

Semester

4

Credits

4

Teaching scheme

4 (L:3,T:1,P:0); 60 periods

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Clutches and transmission systems	15	CO1
2	Driveline, wheels, tyres and alignment	14	CO2
3	Suspension and steering systems	15	CO3
4	Modern brake systems	14	CO4

Prerequisites

- Basic automobile engineering and elementary mechanics.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO-PO Information

Official course objectives

1. Describe the working of various types of clutches and transmission system.
2. Understand the features of driveline components.
3. Describe the various suspension types and steering systems.
4. Explain the working construction of modern brake systems.

Course Outcomes

1. Describe the working of various types of clutches and transmission system.
2. Understand the features of driveline components.
3. Describe the various suspension types and steering systems.
4. Explain the working construction of modern brake systems.

CO-PO mapping as supplied

CO-PO mapping is reproduced from the supplied syllabus. The numerical mapping is retained as printed; blank cells are not silently converted into values.

Legend: 3 = strongly mapped, 2 = moderately mapped, 1 = weakly mapped.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Clutch components and functions	4
module-1	M1.02	Clutch linkages	3
module-1	M1.03	Manual transmission and transaxle	4
module-1	M1.04	Automatic transmission	4
module-2	M2.01	Driveline components and joints	4
module-2	M2.02	Differential	3
module-2	M2.03	Wheels and tyres	3
module-2	M2.04	Wheel-alignment geometry	4
module-3	M3.01	Suspension components	4
module-3	M3.02	Front and rear suspension types	4
module-3	M3.03	Steering components and linkages	4
module-3	M3.04	Power steering systems	3
module-4	M4.01	Hydraulic-brake components	4
module-4	M4.02	Master cylinder and wheel cylinder	3
module-4	M4.03	Drum and disc brakes	4
module-4	M4.04	Power-assisted brakes, air brakes and ABS	3

Learning Roadmap

1. Clutches and transmission systems
CO1 · 15 hours

4. Modern brake systems

2. Driveline, wheels, tyres and alignment
CO2 · 14 hours

3. Suspension and steering systems
CO3 · 15 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Clutches and transmission systems

The clutch connects and disconnects engine torque from the transmission. Transmission and transaxle arrangements then provide suitable speed, torque and direction for vehicle motion.

Hours: 15

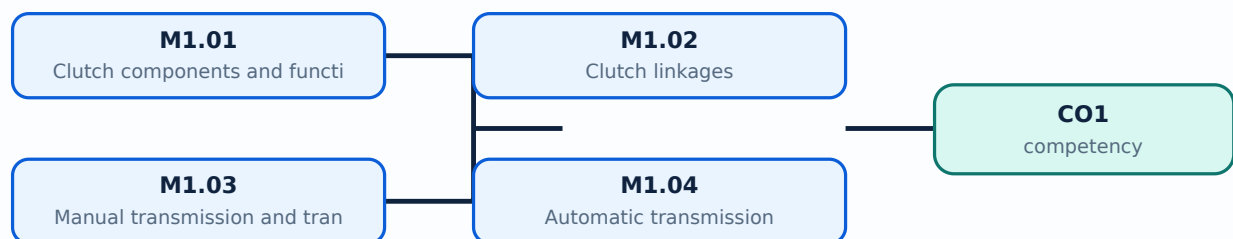
CO1 · Bloom level: Understanding

Learning goals

- Identify clutch parts and functions.
- Compare manual, transaxle and automatic power flow.
- Explain linkages, synchronizers and planetary controls.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

– M1.01 — Clutch components and functions (4 hours)

Concept and explanation

A clutch transmits engine torque when engaged and interrupts torque when disengaged. The clutch disc, pressure plate, release or throw-out bearing, flywheel and pedal linkage work together to provide smooth engagement.

Malayalam support: മലയാളം സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations ഉൾപ്പെടെയുള്ള സാങ്കേതിക പദങ്ങൾ.

Study points

- Name clutch components.
- Explain friction-surface operation.
- Relate pedal travel to release mechanism.

Engineering / workplace application: Automotive drivetrains use clutches during starting, gear changes and controlled stopping.

Illustrative example: When the pedal is pressed, the release bearing acts on the pressure mechanism and separates the friction surfaces.

Common mistake: Confusing the clutch disc with the pressure plate can lead to an incorrect working explanation.

– M1.02 — Clutch linkages (3 hours)

Concept and explanation

Pedal movement reaches the clutch release mechanism through levers and rods, cable linkages or hydraulic linkages. Correct free play is needed to prevent slipping and incomplete release.

Malayalam support: ഏകദേശം ൩ മണിക്കൂർ English technical terms, symbols, units, equations, and standard abbreviations ഉപയോഗിച്ച് വിശദീകരിക്കുക.

Study points

- Compare mechanical cable and hydraulic linkages.
- State the purpose of free play.
- Trace force from pedal to release bearing.

Engineering / workplace application: Linkage adjustment is a routine service issue in vehicles.

Common mistake: Zero free play may keep the clutch partially released; excessive free play may prevent complete disengagement.

– M1.03 — Manual transmission and transaxle (4 hours)

Concept and explanation

A manual transmission selects gear ratios through gears, synchronizers and shift mechanisms. A transaxle combines transmission and final-drive/differential functions, commonly in front-wheel-drive layouts.

Malayalam support: മലയാളം സാങ്കേതിക പദങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations ഇംഗ്ലീഷ് സാങ്കേതിക പദങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations.

Study points

- Explain why gear ratios are required.
- Describe synchronizer action.
- Compare transmission and transaxle power flow.

Engineering / workplace application: Gearboxes allow the engine to operate in an efficient speed range while vehicle speed changes.

Illustrative example: A lower gear provides greater wheel torque for starting or climbing, while a higher gear supports efficient cruising.

Common mistake: A synchronizer equalizes speed before engagement; it is not the same as the clutch.

– M1.04 — Automatic transmission (4 hours)

Concept and explanation

An automatic transmission uses a torque converter, planetary gear sets, hydraulic control and selected shift modes. A dual-clutch automatic uses two clutches to preselect alternate gear paths.

Malayalam support: മലയാളം സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations ഇംഗ്ലീഷ് സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations.

Study points

- Describe torque-converter function.
- Explain planetary gear-set control.
- Identify hydraulic and dual-clutch features.

Engineering / workplace application: Automatic transmissions are used where smooth shifting and reduced driver workload are important.

Common mistake: The torque converter is not a fixed mechanical gear ratio; it transfers torque hydraulically and can multiply torque during launch.

Module summary: Clutch operation, transmission ratios and automatic controls explain how engine power reaches the driving wheels.

OFFICIAL MODULE 2

Driveline, wheels, tyres and alignment

The driveline transfers torque from the transmission to the wheels. Universal joints, constant-velocity joints, shafts, differentials, wheels and tyres must work together while allowing suspension movement.

Hours: 14

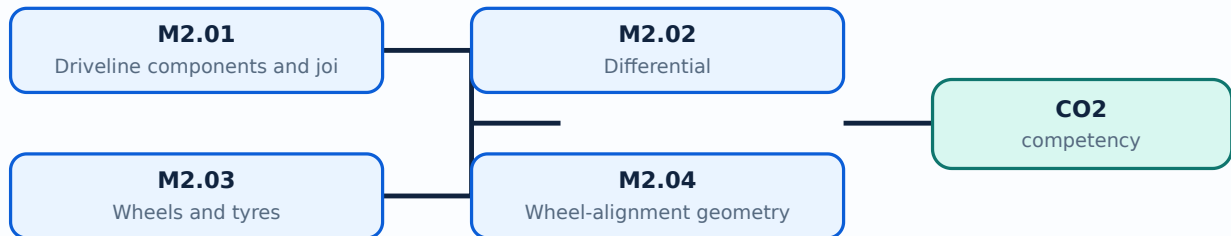
CO2 · Bloom level: Understanding

Learning goals

- Identify driveline components.
- Explain differential operation and four-wheel-drive arrangements.
- Read wheel, tyre and alignment terminology.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

Concept and explanation

The driveline includes universal joints, constant-velocity joints, drive axle shafts, transfer cases and inter-axle differentials. Universal joints accommodate angular change; CV joints maintain smoother constant-speed torque transfer.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- State the purpose of universal and CV joints.
- Identify inner and outer CV joints.
- Explain transfer-case and four-wheel-drive roles.

Engineering / workplace application: Four-wheel-drive and all-wheel-drive vehicles require coordinated torque distribution between axles.

Common mistake: A differential allows speed difference between left and right wheels; it does not automatically provide equal traction.

Concept and explanation

A differential permits the driven wheels to rotate at different speeds during a turn while transmitting torque. Limited-slip differentials reduce excessive one-wheel spin by adding a resistance or locking action.

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Study points

- Name principal differential parts.
- Explain differential action while turning.
- Compare open and limited-slip differentials.

Engineering / workplace application: Differentials are essential because the outer wheel travels a longer path than the inner wheel during a turn.

– M2.03 — Wheels and tyres (3 hours)

Concept and explanation

Wheels support the tyre and transmit braking, driving and steering forces. Tyre construction, tread pattern, rating, designation, pressure monitoring, care and repair affect safety and performance.

Malayalam support: താഴെ പറയുന്നവയെക്കുറിച്ച് English technical terms, symbols, units, equations, and standard abbreviations പരിചയപ്പെടുക.

Study points

- Interpret a tyre designation.
- State tyre-care requirements.
- Relate tread and pressure to grip.

Engineering / workplace application: Tyres are the contact interface between the vehicle and road.

Common mistake: Under-inflation and overloading change heat generation, wear and handling; tyre pressure must follow the specified value.

– M2.04 — Wheel-alignment geometry (4 hours)

Concept and explanation

Caster, camber, toe, steering-axis inclination and turning radius describe wheel orientation and steering geometry. Correct alignment supports directional stability, tyre life and predictable steering.

Malayalam support: മലയാളം സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations ഉപയോഗിക്കുക.

Study points

- Define caster, camber and toe.
- Explain steering-axis inclination.
- Relate misalignment to wear and pull.

Engineering / workplace application: Alignment equipment is used in vehicle service workshops.

Common mistake: Do not treat positive and negative camber as interchangeable signs; draw the wheel geometry before interpreting it.

Module summary: Driveline components transmit torque, while wheels, tyres and alignment determine contact, stability and road response.

OFFICIAL MODULE 3

Suspension and steering systems

Suspension supports the vehicle body, absorbs road irregularities and maintains tyre contact. Steering linkages and power-assist systems convert driver input into controlled wheel direction.

Hours: 15

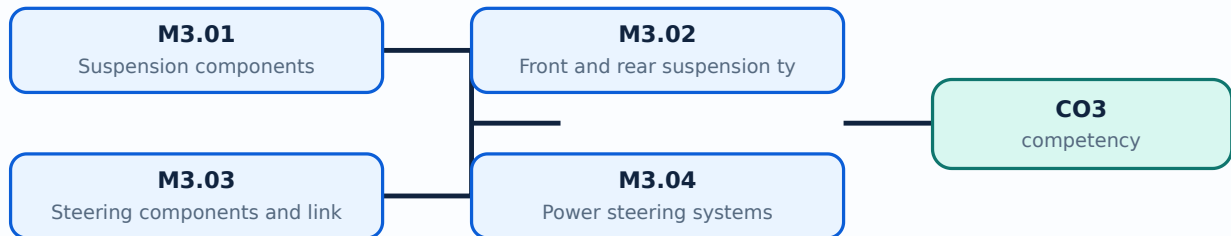
CO3 · Bloom level: Understanding

Learning goals

- Identify suspension and steering components.
- Compare front and rear suspension types.
- Explain mechanical, hydraulic and electronic power steering.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

– M3.01 — Suspension components (4 hours)

Concept and explanation

Steering knuckles, control arms, ball joints, strut rods, stabilizer bars and springs/dampers locate the wheel and control body motion. Suspension principles balance ride comfort, handling and tyre contact.

Malayalam support: ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ English technical terms, symbols, units, equations, and standard abbreviations ഞ ഞ ഞ ഞ ഞ ഞ.

Study points

- State component functions.
- Differentiate spring and damper roles.
- Explain stabilizer-bar action.

Engineering / workplace application: Suspension inspection is central to vehicle safety and ride quality.

– M3.02 — Front and rear suspension types (4 hours)

Concept and explanation

Solid-axle, leaf-spring, trailing-arm and independent suspension arrangements differ in wheel interconnection, load carrying and ride characteristics. Electronic and air suspension add controlled or variable support.

Malayalam support: ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ English technical terms, symbols, units, equations, and standard abbreviations ഞ ഞ ഞ ഞ ഞ ഞ.

Study points

- Compare solid and independent layouts.
- Describe leaf-spring and trailing-arm features.
- Explain air-suspension components and working.

Engineering / workplace application: Commercial vehicles often prioritise load capacity, while passenger vehicles often prioritise ride and handling.

Concept and explanation

Parallelogram and rack-and-pinion linkages transfer steering-wheel movement to the knuckles. Steering gears may use recirculating balls or worm-and-roller arrangements; tilt wheels and collapsible columns improve ergonomics and safety.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- Trace steering motion.
- Compare rack-and-pinion and parallelogram linkages.
- State the purpose of a collapsible column.

Engineering / workplace application: Steering geometry connects directly with wheel alignment and turning radius.

– M3.04 — Power steering systems (3 hours)

Concept and explanation

Power steering reduces driver effort using integral-piston, power-assisted rack-and-pinion or electronic power-steering systems. Assistance must remain predictable and should not hide mechanical defects.

Malayalam support: മലയാളം സാങ്കേതിക പദങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations ഇംഗ്ലീഷ് സാങ്കേതിക പദങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations.

Study points

- Compare hydraulic and electronic assistance.
- Explain assist generation.
- Identify safety implications of loss of assist.

Engineering / workplace application: Modern vehicles use electronic power steering for efficiency and control integration.

Common mistake: Power assistance does not replace steering linkage inspection or correct alignment.

Module summary: Suspension manages vertical motion and steering systems manage lateral direction while preserving tyre contact.

OFFICIAL MODULE 4

Modern brake systems

Braking converts vehicle kinetic energy into heat and must apply controllable force at each wheel. Hydraulic circuits, drum/disc brakes, power assistance, air brakes and ABS provide different levels of control and redundancy.

Hours: 14

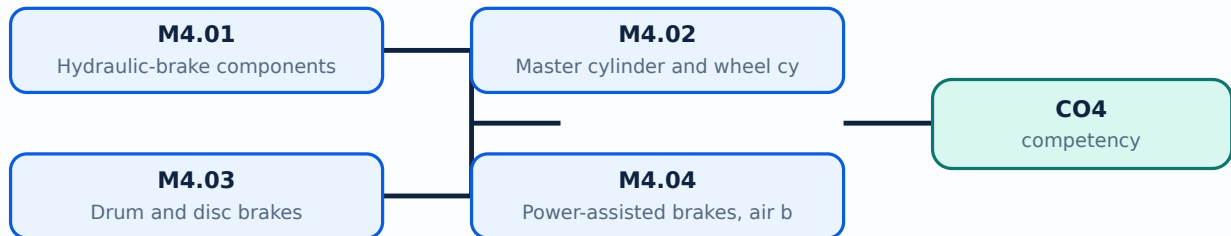
CO4 · Bloom level: Understanding

Learning goals

- Identify hydraulic-brake components.
- Explain master and wheel-cylinder operation.
- Compare drum, disc, assisted and ABS systems.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

Concept and explanation

A hydraulic brake system includes the pedal, master cylinder, hydraulic lines, pressure-differential switch, metering/proportioning valves and wheel actuators. Fluid pressure transmits pedal force to the wheels.

Malayalam support: □ □□□□ □□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- Trace hydraulic pressure.
- State the function of proportioning valves.
- Explain split hydraulic circuits.

Engineering / workplace application: Hydraulic systems are used in passenger and light commercial vehicles.

Common mistake: Air in the hydraulic line is compressible and can produce a spongy pedal; bleeding must follow safe service procedure.

Concept and explanation

The master cylinder converts pedal force into hydraulic pressure. A drum-brake wheel cylinder converts pressure into outward shoe movement; disc-brake calipers convert pressure into pad clamping force.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- Compare master and wheel cylinders.
- Explain piston and seal action.
- Relate pressure to braking force.

Engineering / workplace application: Hydraulic pressure is distributed through a closed circuit with appropriate fluid and seals.

Concept and explanation

Drum brakes use shoes against an internal drum, while disc brakes use pads clamping a rotating disc. Caliper construction and heat dissipation influence performance, wear and fade.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- Compare drum and disc construction.
- Describe caliper types.
- Explain brake fade and heat removal.

Engineering / workplace application: Disc brakes are common where consistent cooling and response are important.

Common mistake: Brake friction surfaces must be clean and correctly bedded; contamination changes braking performance.

– M4.04 — Power-assisted brakes, air brakes and ABS (3 hours)

Concept and explanation

Vacuum and hydraulic boosters reduce pedal effort. Air brakes use compressed air and are common in heavy vehicles. ABS modulates wheel pressure to reduce lock-up and preserve steering control during hard braking.

Malayalam support: മലയാളം സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations ഇംഗ്ലീഷ് സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations.

Study points

- Explain booster operation.
- Identify air-brake components.
- Describe the principle of ABS modulation.

Engineering / workplace application: ABS sensors and hydraulic modulators are integrated with modern vehicle control systems.

Common mistake: ABS does not guarantee a shorter stopping distance on every surface; it mainly maintains controllability during wheel-slip conditions.

Module summary: Effective braking requires force generation, hydraulic or pneumatic transmission, friction, heat management and controlled wheel slip.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

[View its labelled learning-map diagram.](#)

[Module 2: Driveline, wheels,](#)

[to view its labelled learning-map diagram.](#)

[Module 3: Suspension and](#)

[module to view its labelled learning-map diagram.](#)

[Module 4: Modern brake](#)

[Open the module to view its labelled learning-map diagram.](#)

Official Activities and Practical Requirements

Official activities / self-learning

- Study each chassis system using a labelled component sketch.
- Observe safe vehicle-service demonstrations only under instructor supervision.

Practical requirements / equipment

- No separate practical equipment list is provided in the supplied PDF.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- The supplied syllabus specifies Series Test I and Series Test II as 1-hour entries and does not state a separate CIE/ESE mark distribution.

Module-wise Questions and Answers

module-1 — Clutches and transmission systems

1. Explain Clutch components and functions. [3 marks]

A clutch transmits engine torque when engaged and interrupts torque when disengaged. The clutch disc, pressure plate, release or throw-out bearing, flywheel and pedal linkage work together to provide smooth engagement.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Clutch linkages. [3 marks]

Pedal movement reaches the clutch release mechanism through levers and rods, cable linkages or hydraulic linkages. Correct free play is needed to prevent slipping and incomplete release.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Manual transmission and transaxle. [3 marks]

A manual transmission selects gear ratios through gears, synchronizers and shift mechanisms. A transaxle combines transmission and final-drive/differential functions, commonly in front-wheel-drive layouts.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Automatic transmission. [3 marks]

An automatic transmission uses a torque converter, planetary gear sets, hydraulic control and selected shift modes. A dual-clutch automatic uses two clutches to preselect alternate gear paths.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Driveline, wheels, tyres and alignment

5. Explain Driveline components and joints. [3 marks]

The driveline includes universal joints, constant-velocity joints, drive axle shafts, transfer cases and inter-axle differentials. Universal joints accommodate angular change; CV joints maintain smoother constant-speed torque transfer.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Differential. [3 marks]

A differential permits the driven wheels to rotate at different speeds during a turn while transmitting torque. Limited-slip differentials reduce excessive one-wheel spin by adding a resistance or locking action.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Wheels and tyres. [3 marks]

Wheels support the tyre and transmit braking, driving and steering forces. Tyre construction, tread pattern, rating, designation, pressure monitoring, care and repair affect safety and performance.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Wheel-alignment geometry. [3 marks]

Caster, camber, toe, steering-axis inclination and turning radius describe wheel orientation and steering geometry. Correct alignment supports directional stability, tyre life and predictable steering.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Suspension and steering systems

9. Explain Suspension components. [3 marks]

Steering knuckles, control arms, ball joints, strut rods, stabilizer bars and springs/dampers locate the wheel and control body motion. Suspension principles balance ride comfort, handling and tyre contact.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Front and rear suspension types. [3 marks]

Solid-axle, leaf-spring, trailing-arm and independent suspension arrangements differ in wheel interconnection, load carrying and ride characteristics. Electronic and air suspension add controlled or variable support.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Steering components and linkages. [3 marks]

Parallelogram and rack-and-pinion linkages transfer steering-wheel movement to the knuckles. Steering gears may use recirculating balls or worm-and-roller arrangements; tilt wheels and collapsible columns improve ergonomics and safety.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Power steering systems. [3 marks]

Power steering reduces driver effort using integral-piston, power-assisted rack-and-pinion or electronic power-steering systems. Assistance must remain predictable and should not hide mechanical defects.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Modern brake systems

13. Explain Hydraulic-brake components. [3 marks]

A hydraulic brake system includes the pedal, master cylinder, hydraulic lines, pressure-differential switch, metering/proportioning valves and wheel actuators. Fluid pressure transmits pedal force to the wheels.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Master cylinder and wheel cylinder. [3 marks]

The master cylinder converts pedal force into hydraulic pressure. A drum-brake wheel cylinder converts pressure into outward shoe movement; disc-brake calipers convert pressure into pad clamping force.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Drum and disc brakes. [3 marks]

Drum brakes use shoes against an internal drum, while disc brakes use pads clamping a rotating disc. Caliper construction and heat dissipation influence performance, wear and fade.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Power-assisted brakes, air brakes and ABS. [3 marks]

Vacuum and hydraulic boosters reduce pedal effort. Air brakes use compressed air and are common in heavy vehicles. ABS modulates wheel pressure to reduce lock-up and preserve steering control during hard braking.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Clutch components and functions* with one relevant application. (5 marks)
2. **2.** Define or explain *Clutch linkages* with one relevant application. (5 marks)
3. **3.** Define or explain *Driveline components and joints* with one relevant application. (5 marks)
4. **4.** Define or explain *Differential* with one relevant application. (5 marks)
5. **5.** Define or explain *Suspension components* with one relevant application. (5 marks)
6. **6.** Define or explain *Front and rear suspension types* with one relevant application. (5 marks)
7. **7.** Define or explain *Hydraulic-brake components* with one relevant application. (5 marks)
8. **8.** Define or explain *Master cylinder and wheel cylinder* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Clutches and transmission systems:** Clutch operation, transmission ratios and automatic controls explain how engine power reaches the driving wheels.
- **Driveline, wheels, tyres and alignment:** Driveline components transmit torque, while wheels, tyres and alignment determine contact, stability and road response.
- **Suspension and steering systems:** Suspension manages vertical motion and steering systems manage lateral direction while preserving tyre contact.
- **Modern brake systems:** Effective braking requires force generation, hydraulic or pneumatic transmission, friction, heat management and controlled wheel slip.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Clutch components and functions	A clutch transmits engine torque when engaged and interrupts torque when disengaged. The clutch disc, pressure plate, release or throw-out bearing, flywheel and pedal linkage work together to provide smooth engagement.
Clutch linkages	Pedal movement reaches the clutch release mechanism through levers and rods, cable linkages or hydraulic linkages. Correct free play is needed to prevent slipping and incomplete release.
Manual transmission and transaxle	A manual transmission selects gear ratios through gears, synchronizers and shift mechanisms. A transaxle combines transmission and final-drive/differential functions, commonly in front-wheel-drive layouts.
Automatic transmission	An automatic transmission uses a torque converter, planetary gear sets, hydraulic control and selected shift modes. A dual-clutch automatic uses two clutches to preselect alternate gear paths.
Driveline components and joints	The driveline includes universal joints, constant-velocity joints, drive axle shafts, transfer cases and inter-axle differentials. Universal joints accommodate angular change; CV joints maintain smoother constant-speed torque transfer.
Differential	A differential permits the driven wheels to rotate at different speeds during a turn while transmitting torque. Limited-slip differentials reduce excessive one-wheel spin by adding a resistance or locking action.
Wheels and tyres	Wheels support the tyre and transmit braking, driving and steering forces. Tyre construction, tread pattern, rating, designation, pressure monitoring, care and repair affect safety and performance.
Wheel-alignment geometry	Caster, camber, toe, steering-axis inclination and turning radius describe wheel orientation and steering geometry. Correct alignment supports directional stability, tyre life and predictable steering.

Term	Short meaning
Suspension components	Steering knuckles, control arms, ball joints, strut rods, stabilizer bars and springs/dampers locate the wheel and control body motion. Suspension principles balance ride comfort, handling and tyre contact.
Front and rear suspension types	Solid-axle, leaf-spring, trailing-arm and independent suspension arrangements differ in wheel interconnection, load carrying and ride characteristics. Electronic and air suspension add controlled or variable support.
Steering components and linkages	Parallelogram and rack-and-pinion linkages transfer steering-wheel movement to the knuckles. Steering gears may use recirculating balls or worm-and-roller arrangements; tilt wheels and collapsible columns improve ergonomics and safety.
Power steering systems	Power steering reduces driver effort using integral-piston, power-assisted rack-and-pinion or electronic power-steering systems. Assistance must remain predictable and should not hide mechanical defects.
Hydraulic-brake components	A hydraulic brake system includes the pedal, master cylinder, hydraulic lines, pressure-differential switch, metering/proportioning valves and wheel actuators. Fluid pressure transmits pedal force to the wheels.
Master cylinder and wheel cylinder	The master cylinder converts pedal force into hydraulic pressure. A drum-brake wheel cylinder converts pressure into outward shoe movement; disc-brake calipers convert pressure into pad clamping force.
Drum and disc brakes	Drum brakes use shoes against an internal drum, while disc brakes use pads clamping a rotating disc. Caliper construction and heat dissipation influence performance, wear and fade.
Power-assisted brakes, air brakes and ABS	Vacuum and hydraulic boosters reduce pedal effort. Air brakes use compressed air and are common in heavy vehicles. ABS modulates wheel pressure to reduce lock-up and preserve steering control during hard braking.

Official References and Resources

References listed in the supplied syllabus

1. Automotive Chassis and Body, J. Heitner.
2. Automotive Mechanics, William H. Crouse and Donald L. Anglin.

Official learning resources listed

- <https://nptel.ac.in/>

In-page Search

